

BOLTZMANN-REGULARIZED FLOW MATCHING FOR UNIVERSAL-NEURAL-POTENTIAL GUIDED 3D MOLECULAR GENERATION

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ABSTRACT

Modern de novo 3D molecular generators can produce valid, data-like structures, but their learned coordinate densities are not explicitly tied to a physically meaningful energy landscape. As a result, generated geometries may require substantial downstream relaxation before use in simulation or screening. We propose *Boltzmann-Regularized Flow Matching* (BGFM), a training-time physical regularization framework for 3D molecular flow matching. BGFM keeps a FlowMol3-style de novo generator as the backbone, but uses energies and forces from a universal neural potential, OMol25, to align the model’s coordinate density with local conditional Boltzmann relative probabilities. For a fixed molecular identity c and coordinates r , OMol25 induces the target

$$p_B(r \mid c, T) = Z_c(T)^{-1} \exp[-E_{\text{NP}}(r, c)/kT].$$

BGFM does not claim to sample an exact global Boltzmann distribution over arbitrary atom counts, compositions, and charges. Instead, it regularizes the position-only conditional density $p_\theta^{\text{pos}}(r \mid c)$ using two complementary signals: a late-time force-score consistency loss and a FFJORD density-energy variance loss over pre-computed perturbation clouds. A small composition-conditioned anchor stabilizes the otherwise unidentified additive density offset. This turns a universal neural potential from a post-generation relaxation tool into a train-time supervisor for de novo molecular generation. We evaluate BGFM by local Boltzmann correlation, force-score alignment, downstream relaxation cost, and ensemble overlap, in addition to conventional validity metrics.

1 INTRODUCTION

Three-dimensional molecular generation is usually trained as a data-imitation problem: a model learns to generate molecular graphs and coordinates that resemble examples in a training set. This objective is necessary but incomplete. A generated molecule is not merely an image-like object; its coordinates should lie on a physically meaningful energy landscape. A structure that is formally valid but high in energy, sterically strained, or poorly aligned with a neural-potential force field may be expensive to relax and unreliable for downstream simulation.

Universal neural potentials create a new opportunity. Rather than using a potential only after generation to relax or filter samples, we can use it during training to shape the generator itself. BGFM follows this idea. Given a FlowMol3-style flow-matching generator (Stark et al., 2024) and an OMol25 neural potential (Levine et al., 2025), BGFM adds physical supervision so that local coordinate probabilities become consistent with OMol25 energy differences and forces.

The central distinction is important: BGFM is not a new neural architecture and not an exact Boltzmann generator over all molecular identities. It is a *physical regularization layer for de novo 3D flow matching*. The method keeps the backbone generator, but changes the training target and wraps the conditional path and sampling execution with geometric constraint hooks.

Contributions.

1. We formulate de novo 3D molecular generation as *conditional coordinate Boltzmann alignment*: for fixed molecular identity c , the generated coordinate density should locally satisfy $\log p_{\theta}^{\text{pos}}(r | c) + E_{\text{NP}}(r, c)/kT \approx \text{const}(c, T)$.
2. We derive a late-time force-score regularizer for flow matching, comparing the FM-implied score at path states x_t with endpoint OMol25 forces $F_{\text{NP}}(x_1)/kT$ (§3.3).
3. We introduce a position-only FFJORD density-energy variance loss over per-molecule perturbation clouds, which enforces local Boltzmann relative probabilities without requiring the true partition function Z_c (§3.4).
4. We add a composition-conditioned offset anchor that stabilizes the absolute density offset while preserving permutation, translation, and rotation invariance (§3.5).
5. We propose physical evaluation metrics for de novo generators: local Boltzmann correlation, energy-log-density slope, force-score cosine, downstream relaxation cost, and ensemble overlap (§4).

2 RELATED WORK

3D molecular generation. EDM (Hoogeboom et al., 2022), GeoDiff (Xu et al., 2022), MiDi (Vignac et al., 2023), GeoLDM (Xu et al., 2023), GeoBFN (Song et al., 2024), FlowMol/FlowMol3 (Stark et al., 2024), and JoMo (Le et al., 2024) train sample-set quality losses (validity, FCD); none enforce Boltzmann. Torsional Diffusion (Jing et al., 2022) restricts to torsional manifolds for conformer generation given a 2D graph. CGFlow (Tang et al., 2025) integrates GFlowNet reward-guided sampling with a discrete-step FM formulation for synthesizable drug design.

Boltzmann generators. Boltzmann Generators (Noé et al., 2019), Equivariant FM (Klein et al., 2023), Implicit Transfer Operator Learning (Schreiner et al., 2023), and Scalable Boltzmann Generators (Wang et al., 2025) target Boltzmann densities, but for *fixed* classical potentials on *single* systems. The Adjoint Schrödinger Bridge Sampler (Liu et al., 2025) amortizes Boltzmann sampling across $\sim 24\text{K}$ SPICE topologies, but only over conformer generation conditional on the 2D graph, and only over ~ 10 -element drug-like chemistry. BGFM differs in two ways: the target distribution is the position-only conditional density $p_{\theta}(r | c)$ across de novo identities c , and the energy oracle is a *universal* neural potential covering 83 elements.

Energy-based molecular generation (closest competitor). EBMol (Anonymous, 2025) trains a time-unconditional EBM directly on QM9 and GEOM-Drugs and samples it with Mirror-Langevin plus parallel tempering. EBMol’s strongest claim is that an explicit, learned scalar energy is more useful than a denoising-only objective for ranking, filtering, steering, and zero-shot linker design. BGFM addresses the same underlying problem (sampler quality coupled to energy structure) through a complementary route: keep an amortized flow generator and supervise it with an *externally calibrated* universal neural potential. We compare against EBMol on its own QM9 and GEOM-Drugs benchmarks (Sections 4.2–??), on independent GFN2-xTB physical metrics (Section 4.3), on quality-diversity Pareto frontiers, and on cross-model energy ranking (Section 4.7). The two methods occupy different points in the design space rather than being mutually exclusive: an extension of BGFM that adds a calibrated scalar energy head on top of the flow generator (denoted BGFM++ in Section 5) is left to future work.

Universal neural potentials. SchNet (Schütt et al., 2018), NequIP (Batzner et al., 2022), MACE-MP-0 (Battat et al., 2023), and OMol25/eSEN (Levine et al., 2025) act as DFT replacements for molecular dynamics and relaxation, but have not been coupled into a generative training loop at scale. BGFM is the first to embed a universal NP as a supervision signal inside the generative training objective.

FFJORD-style density inside training. Grathwohl et al. (2019) and Chen et al. (2018) compute log-density for continuous normalizing flows via the instantaneous change-of-variables formula. Standard flow matching (Lipman et al., 2023) explicitly avoids log-density to bypass the divergence integral. BGFM revisits this trade-off because Boltzmann targets require global log-density alignment; we show the cost can be controlled through a per-molecule variance form and step-amortized scheduling.

3 METHOD

3.1 CONDITIONAL BOLTZMANN TARGET

Let c denote the discrete molecular identity: atom count, atom types, total charge, and any fixed graph/discrete state used by the generator. Let $r \in \mathbb{R}^{3N}$ denote coordinates. BGFM does not define a grand-canonical distribution over c . Instead, for each fixed c , a neural potential $E_{\text{NP}}(r, c)$ induces a coordinate Boltzmann density

$$p_B(r | c, T) = \frac{1}{Z_c(T)} \exp[-\beta E_{\text{NP}}(r, c)], \quad \beta = (kT)^{-1}. \quad (1)$$

The goal is to regularize the generator’s coordinate density $p_\theta^{\text{pos}}(r | c)$ so that local relative probabilities match this target near chemically relevant geometries.

3.2 SCORE FROM FLOW-MATCHING VELOCITY

For the Gaussian-prior linear interpolant $x_t = (1-t)x_0 + tx_1$ with $x_0 \sim \mathcal{N}(0, \sigma^2 I)$, a flow-matching velocity field $v_\theta(x_t, t)$ implies a marginal score approximation

$$s_\theta(x_t, t) = \frac{tv_\theta(x_t, t) - x_t}{(1-t)\sigma^2}. \quad (2)$$

The expression is singular at $t = 1$, so BGFM evaluates it at late but finite path times $t \in \{0.70, 0.80, 0.90\}$. A self-contained derivation is given in Appendix A.1.

3.3 LATE-TIME FORCE-SCORE CONSISTENCY

For a Boltzmann coordinate density, $\nabla_r \log p_B(r | c, T) = F_{\text{NP}}(r, c)/kT$, where $F_{\text{NP}} = -\nabla_r E_{\text{NP}}$. BGFM therefore regularizes the FM-implied score toward OMol25 forces:

$$\mathcal{L}_{\text{force}} = \mathbb{E}_{m, t_{\text{eval}}} [1 - \cos(s_\theta(x_t^m, t_{\text{eval}}), F_{\text{NP}}(x_1^m, c_m)/kT)]. \quad (3)$$

The current implementation pairs the score evaluated at path state x_t^m on the FM conditional path with the precomputed endpoint force $F_{\text{NP}}(x_1^m)$. This is a *late-time approximation* $F_{\text{NP}}(x_t) \approx F_{\text{NP}}(x_1)$, used to avoid querying OMol25 inside every training step; at $t_{\text{eval}} \in \{0.7, 0.8, 0.9\}$ the discrepancy is first-order in $\|x_t - x_1\|$ and the cosine variant of the loss is robust to magnitude mismatch.

3.4 DENSITY-ENERGY VARIANCE OVER PERTURBATION CLOUDS

Force alignment constrains local gradients but not the scalar density-energy relationship. BGFM therefore estimates a position-only log-density using a FFJORD-style divergence integral (Chen et al., 2018; Grathwohl et al., 2019) while holding atom types and charges fixed. For perturbations $r_m^{(k)}$ of the same parent molecule m , define

$$y_m^{(k)} = \log p_\theta^{\text{pos}}(r_m^{(k)} | c_m) + \beta E_{\text{NP}}(r_m^{(k)}, c_m).$$

If p_θ^{pos} matches the conditional Boltzmann density within the local cloud of m , then $y_m^{(k)}$ should be constant across k (equal to $-\log Z_{c_m}$). BGFM minimizes

$$\mathcal{L}_{\text{energy}} = \mathbb{E}_m \text{Var}_k [y_m^{(k)}]. \quad (4)$$

This loss enforces relative Boltzmann probabilities without needing the unknown partition function Z_{c_m} . The variance form is the key mathematical observation: variance is translation-invariant in k , so the loss is invariant to any c -only shift, including the unknown $-\log Z_{c_m}$ itself (Appendix A.2).

The FFJORD log-density $\log p_\theta^{\text{pos}}(r_m^{(k)} | c_m)$ is evaluated via the instantaneous change-of-variables formula (Chen et al., 2018)

$$\log p_\theta^{\text{pos}}(x_1) = \log p_0(x_0) - \int_0^1 \nabla_x \cdot v_\theta(x_t, t) dt, \quad (5)$$

with $\nabla_x \cdot v_\theta$ estimated by Hutchinson’s trick (Hutchinson, 1990). To control cost we use 4 ODE steps and 4 Hutchinson samples, and amortize by evaluating $\mathcal{L}_{\text{energy}}$ on every fourth training step.

3.5 COMPOSITION-CONDITIONED OFFSET ANCHOR

The variance objective equation 4 is insensitive to additive density offsets. We add a small invariant network $a_\phi(c)$, computed from atom-type counts and total charge, and penalize

$$\mathcal{L}_{\text{anchor}} = \mathbb{E}_{m,k} \left(\log p_\theta^{\text{pos}}(r_m^{(k)} | c_m) + \beta E_{\text{NP}}(r_m^{(k)}, c_m) + a_\phi(c_m) \right)^2. \quad (6)$$

We interpret a_ϕ as a *composition-conditioned offset stabilizer*, not as a supervised or guaranteed estimator of the true thermodynamic partition function $Z_c(T)$. Because a_ϕ takes only composition as input, it cannot encode geometric structure and acts as a per-molecule constant.

3.6 TOTAL OBJECTIVE AND TRAINING SCHEDULE

The final objective is

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda_1(t_{\text{step}}) \mathcal{L}_{\text{force}} + \lambda_2(t_{\text{step}}) \mathcal{L}_{\text{energy}} + \lambda_3(t_{\text{step}}) \mathcal{L}_{\text{anchor}}, \quad (7)$$

where $\lambda_i(t_{\text{step}})$ ramps each BGFM term in linearly from 0 during the first 0.5% of training (warm-up), to a target peak value over the next 2% (ramp), and remains constant thereafter. This protects the FM reconstruction from being destabilized by the BGFM terms before v_θ has a sensible geometric meaning. Target peak values at $kT = 0.025$ eV (room temperature) are $\lambda_1 = 0.05$, $\lambda_2 = 10^{-4}$ (scaled 1/40 relative to a $kT = 1$ eV baseline because E/kT scales as $1/kT$), and $\lambda_3 = 10^{-10}$ (the anchor magnitude $|\log p + E/kT|^2$ can reach 10^{10} – 10^{11} at room temperature; λ_3 keeps its contribution comparable to $\mathcal{L}_{\text{FM}}$).

3.7 OFFLINE PERTURBATION PRECOMPUTE

For each parent molecule m in the training shard, we precompute $K=5$ perturbations at displacement standard deviations $\sigma_p \in \{0.03, 0.06, 0.10, 0.20, 0.40\}$ Å. For each $r_m^{(k)}$ we evaluate OMol25 to obtain $E_{\text{NP}}(r_m^{(k)}, c_m)$ and $F_{\text{NP}}(r_m^{(k)}, c_m)$. The shards are written once to disk and reused across runs, decoupling OMol25 from the training inner loop.

3.8 OPTIONAL TEMPERATURE CONDITIONING

To enable a single trained model to sample at multiple temperatures, we introduce kT as a conditioning variable via a small projection $\pi_\phi(\log kT)$ added residually to the scalar feature stream after the FlowMol3 scalar-embedding block. The projection is zero-initialized, so the temperature-conditional model is exactly equivalent to the base model at initialization and can warm-start from any non-conditional checkpoint. During training, $\log kT$ is sampled log-uniform per step over $[0.025, 1.0]$ eV.

4 EXPERIMENTS

Experimental questions. Our experiments are designed to answer five questions. First, does BGFM preserve the standard chemical validity and diversity expected from de novo 3D molecular generators? Second, does BGFM improve physical quality under independent evaluators, rather than only under the OMol25 neural potential used for training? Third, does the proposed objective actually align conditional coordinate densities with local Boltzmann relative probabilities? Fourth, are the gains caused by correct energy and force information rather than by arbitrary regularization or evaluation leakage? Finally, compared with energy-based samplers such as EBMol, does BGFM offer a better quality–diversity–compute frontier by retaining an amortized flow generator while injecting external neural-potential supervision?

Evaluation philosophy. Since BGFM is trained with OMol25-derived energies and forces, OMol25-based correlation is treated as a mechanism diagnostic rather than the primary external benchmark. Our headline physical-quality results use independent evaluators: GFN2-xTB relaxation (Bannwarth et al., 2019), MMFF (Halgren, 1996) relaxation on organic subsets, and a small DFT validation subset. This separates three claims that are often conflated in physically guided generation: standard generative quality, transferable physical quality, and mechanistic Boltzmann alignment.

Comparison to EBMol. EBMol (Anonymous, 2025) learns a persistent scalar energy landscape and samples it with Mirror-Langevin dynamics and parallel tempering. BGFM takes a complementary route: it keeps an amortized FlowMol3-style generator, but regularizes its conditional coordinate density using an external universal neural potential. We therefore compare not only final validity and stability, but also the quality–diversity–compute frontier: the number of valid, connected, unique, low-strain molecules produced per GPU-hour.

Avoiding self-fulfilling evaluation. To test whether improvements come from correct physical information rather than arbitrary loss shaping, we include shuffled-force, shuffled-energy, wrong-parent-grouping, wrong-temperature, force-norm-only, and random-regularizer controls. A valid BGFM improvement should transfer to independent xTB/MMFF/DFT metrics and should disappear or weaken under these negative controls.

4.1 BENCHMARKS AND BASELINES

We evaluate on three regimes. The first is the standard QM9 benchmark, where we report atom stability, molecule stability, validity, uniqueness, valid-and-unique rate, and novelty on 10,000 generated molecules. The second is GEOM-Drugs (Axelrod & Gómez-Bombarelli, 2022), where we follow the revised evaluation protocol used by recent physically grounded 3D generation work: bond inference is performed with OpenBabel, formal charge and aromaticity are inferred prior to RDKit sanitization, and diversity is measured with the Vendi score (Friedman & Dieng, 2023). The third is an OMol25-based broad-chemistry benchmark, where we evaluate held-out slices by charge, atom count, element composition, conformational flexibility, and, when supported by the evaluator, metal-containing or electrolyte-like species.

We compare against four classes of baselines. *Data references* and *noise references* provide sanity checks. *Standard 3D generative baselines* include EDM (Hooeboom et al., 2022), GeoLDM (Xu et al., 2023), GeoBFN (Song et al., 2024), SLDM/SemlaFlow when checkpoints are available, and FlowMol3 (Stark et al., 2024). EBMol (Anonymous, 2025) is our primary energy-based competitor. Finally, we include *controlled BGFM variants* built on the same FlowMol3-style backbone: FM without BGFM, BGFM-force, BGFM-energy, BGFM-full, and BGFM-T. When reporting an optional refinement variant, BGFM++ denotes BGFM with a calibrated scalar energy head and a short learned-energy Langevin or MALA correction stage (Appendix D); all oracle calls, refinement steps, NFEs, and wall-clock costs are reported.

4.2 EXPERIMENT 1: STANDARD UNCONDITIONAL GENERATION

Goal. Show BGFM does not sacrifice ordinary generative quality.

Protocol. For each model, we generate 10,000 molecules using the same atom-count distribution as the corresponding benchmark. On QM9 we follow the standard EDM-style protocol; on GEOM-Drugs we use the revised OpenBabel/RDKit protocol. Tables include model parameters, NFEs, wall-clock time, post-processing steps, and the number of failed samples.

EBMol’s strongest QM9 row (NFE = 1810) reports atom stability 99.76%, molecule stability 97.09%, validity 98.93%, uniqueness 82.30%, novelty 42.08%. EBMol’s GEOM-Drugs row at NFE = 7240 reports diversity 645.0 vs. data diversity 901.7. Our target claim is therefore the Pareto frontier rather than absolute peak stability: matched stability with substantially higher uniqueness, novelty, and diversity.

4.3 EXPERIMENT 2: INDEPENDENT PHYSICAL QUALITY

Goal. Directly answer the criticism “you trained on OMol25 energy/force, so OMol25 evaluation is biased.”

Protocol. Generate 10,000 samples per model. For each generated molecule we run GFN2-xTB (Bannwarth et al., 2019) relaxation and record the energy decrease ΔE_{relax} , the initial force norm, the number of optimization steps, the convergence failure rate, and the RMSD after relaxation. On organic subsets we additionally report MMFF94 (Halgren, 1996) relaxation statistics. For a

Table 1: Standard unconditional generation metrics on QM9 and GEOM-Drugs (revised OpenBabel/RDKit protocol). The key comparison is not maximum stability alone, but the quality–diversity–compute frontier. Numbers will be filled when EBMol checkpoints are released and BGFM v8/v9 evaluations complete.

Model	NFE	Params	Atom $\uparrow$	Mol $\uparrow$	Valid $\uparrow$	Unique $\uparrow$	Novel $\uparrow$	Div $\uparrow$
EDM	1000	–	–	–	–	–	–	–
GeoLDM	1000	–	–	–	–	–	–	–
GeoBFN	1000	–	–	–	–	–	–	–
FlowMol3	matched	–	–	–	–	–	–	–
EBMol-low	930/1080	–	–	–	–	–	–	–
EBMol-high	1810/7240	–	–	–	–	–	–	–
FM backbone	matched	–	–	–	–	–	–	–
BGFM-force	matched	–	–	–	–	–	–	–
BGFM-energy	matched	–	–	–	–	–	–	–
BGFM-full	matched	–	–	–	–	–	–	–
BGFM++	matched	–	–	–	–	–	–	–

Table 2: Independent physical quality on GEOM-Drugs. The primary quantities are computed using evaluators not identical to the OMol25 labels used during BGFM training. EBMol’s reference numbers (NFE = 7240) are median xTB $\Delta E = 1.78$ kcal/mol, mean 5.27.

Model	V-c $\uparrow$	med $\Delta E \downarrow$	mean $\Delta E \downarrow$	p90 $\Delta E \downarrow$	$\ F\ \downarrow$	Steps $\downarrow$	Fail $\downarrow$	Div $\uparrow$
EBMol-low	–	–	–	–	–	–	–	–
EBMol-high	–	–	–	–	–	–	–	–
FlowMol3	–	–	–	–	–	–	–	–
FM backbone	–	–	–	–	–	–	–	–
BGFM-force	–	–	–	–	–	–	–	–
BGFM-energy	–	–	–	–	–	–	–	–
BGFM-full	–	–	–	–	–	–	–	–
BGFM++	–	–	–	–	–	–	–	–

random subset of 200–500 valid-and-connected molecules per model, we perform DFT single-point calculations or short DFT relaxations as a high-cost validation set.

We report two versions of each metric: a **valid-connected-only** protocol that matches prior structural-energy tables, and an **all-samples** protocol that counts invalid, disconnected, or failed molecules explicitly. The all-samples version prevents models from appearing physically strong only after aggressive filtering.

4.4 EXPERIMENT 3: QUALITY–DIVERSITY–COMPUTE FRONTIER

Goal. Translate EBMol’s mode-concentration trade-off (its GEOM-Drugs diversity drops from 1096.8 at NFE 1080 to 645.0 at NFE 7240) into a benchmark BGFM is well-positioned to win.

Protocol. We compare methods at multiple compute budgets. For EBMol we evaluate the reported low-, medium-, and high-NFE samplers. For BGFM we report matched ODE budgets and matched wall-clock budgets; for BGFM++ we additionally sweep the number of learned-energy refinement steps. We plot molecule stability, median xTB ΔE_{relax} , and valid-connected-low-energy throughput against Vendi diversity, NFEs, and wall-clock time.

We define a throughput metric

$$\text{VLU}_{\tau} = \frac{\#\{\text{valid, connected, unique samples with } \Delta E_{\text{xTB}} < \tau\}}{\text{GPU-hours}}, \quad (8)$$

for thresholds $\tau \in \{2, 5, 10\}$ kcal/mol. This measures how many chemically valid, physically low-strain, non-duplicate molecules a method delivers per fixed computational budget.

Table 3: Ablations and negative controls. Correct energy and force labels should improve both the OMol25 mechanism metric and independent xTB/DFT physical quality; shuffled-label controls should not.

Model/control	OMol25 corr. $\uparrow$	xTB med ΔE $\downarrow$	DFT ΔE $\downarrow$	Valid $\uparrow$	Div $\uparrow$
FM backbone	—	—	—	—	—
BGFM true labels	—	—	—	—	—
BGFM-force only	—	—	—	—	—
BGFM-energy only	—	—	—	—	—
Shuffled force	—	—	—	—	—
Shuffled energy	—	—	—	—	—
Wrong parent grouping	—	—	—	—	—
Wrong kT	—	—	—	—	—
Random regularizer	—	—	—	—	—

4.5 EXPERIMENT 4: MECHANISTIC BOLTZMANN CONSISTENCY

Goal. *Mechanism diagnostic only*, not the main external benchmark.

Protocol. For held-out parent molecules, we construct perturbation clouds using perturbation families not identical to the training shards: unseen Gaussian noise levels, torsion rotations, bond stretches, angle bends, RDKit conformer perturbations, and short xTB or OMol25 Langevin perturbations. For each perturbation $r_m^{(k)}$ of parent m we estimate $\log p_\theta^{\text{pos}}(r_m^{(k)} | c_m)$ and compare to $-E_{\text{oracle}}(r_m^{(k)}, c_m)/kT$.

We report within-parent centered Pearson correlation, Spearman correlation, the slope of $\log p_\theta^{\text{pos}}$ vs. $-E/kT$, the slope calibration error, and the residual variance, separately for three oracle tiers: **OMol25** (the direct mechanism diagnostic), **GFN2-xTB** (independent), and a **DFT subset** (high-cost validation).

Current verified result. BGFM force-only at training step 50,000 reaches within-molecule $R^2 = 0.278$ overall vs. 0.091 for the FM baseline (a $3.0\times$ improvement), and $R^2 = 0.501$ on transition-metal molecules, under the post-2026-06-29 bug-fixed eval pipeline (Appendix C). *This number verifies that BGFM has implemented its intended target; the headline claims are supported by Experiments 4.2–4.4 and the cross-oracle generalization within this experiment.*

4.6 EXPERIMENT 5: ABLATIONS AND NEGATIVE CONTROLS

Goal. Establish causality. The headline story is that real BGFM lifts the independent xTB / DFT metrics; shuffled-label controls leave these metrics unchanged even if they sometimes lift the training-correlated metric.

We train and evaluate *architecture ablations* (FM without BGFM, BGFM-force, BGFM-energy, BGFM-full, BGFM-T) and *negative controls* (shuffled forces across atoms, shuffled forces across molecules, force-norm-only supervision, shuffled energies within perturbation groups, shuffled parent grouping, wrong- kT evaluation, and a random scalar regularizer with the same weight schedule).

4.7 EXPERIMENT 6: ENERGY-BASED RANKING AND FILTERING

Goal. Beat EBMol on its central capability claim: that an explicit scalar energy predicts physical strain on samples generated by other models.

Protocol. We construct a cross-model pool containing samples from EDM, GeoLDM, FlowMol3, EBMol, FM, BGFM, deliberately strained perturbations, and relaxed reference structures. We score each molecule using the EBMol learned energy, the OMol25 energy, the BGFM density-energy score

$$S_{\text{BGFM}}(x) = \log p_\theta^{\text{pos}}(r | c) - \beta E_{\text{OMol25}}(r, c) - b_\phi(c), \quad (9)$$

and, when available, the BGFM++ calibrated scalar energy head. Ground truth is GFN2-xTB relaxation energy, xTB force norms, and DFT subset energy decrease. Metrics are Pearson, Spearman,

AUROC for identifying the top 10% high-strain molecules, and the median relaxation energy after keeping the top-scored 10%, 20%, and 50% of samples.

The target is to beat EBMol’s reported Pearson $r = 0.65$ vs. xTB relaxation energy decrease, using a scorer whose components are fully explicit.

4.8 EXPERIMENT 7: BROAD-CHEMISTRY OMOL25 GENERALIZATION

Goal. BGFM’s universal-NP training opens chemistry coverage that QM9 and GEOM-Drugs do not exercise.

Protocol. We evaluate BGFM on held-out OMol25 chemistry slices stressing broad element coverage, charge states, molecular size, and coordination environments. Results are reported separately for (i) neutral vs. charged molecules, (ii) small vs. large atom counts, (iii) heteroatom-rich molecules, (iv) flexible molecules, (v) rare-element cases, and (vi) metal-containing or electrolyte-like systems where the evaluator supports them. The main comparison is between the FlowMol3-style FM backbone and BGFM variants; if an EBMol-style model is retrained on an OMol25 subset, it is clearly labeled as our reimplement.

Summary. This evaluation separates three claims that should not be conflated: standard de novo generation quality, transferable physical quality under independent oracles, and the mechanistic Boltzmann alignment induced by BGFM. OMol25-based Boltzmann correlation (Experiment 4.5) is used to verify the proposed mechanism, while the headline comparison to EBMol is based on standard QM9/GEOM metrics (Experiment 4.2), independent physical relaxation metrics (Experiment 4.3), and quality–diversity–compute Pareto curves (Experiment 4.4).

5 LIMITATIONS AND BGFM++ EXTENSION

Limitations. BGFM regularizes a position-only conditional density, not the full joint distribution over arbitrary molecular identities. The force term currently compares scores at late path states to endpoint forces, which is a first-order approximation. The composition-conditioned offset anchor is not a true partition-function estimator. The method inherits OMol25 coverage and accuracy limits, especially for rare charge/spin/element regimes. Stable full-loss training and charged-species evaluation must be verified before making strong universal-deployment claims. Finally, our Tier 0 metric (OMol25 log-density vs. energy on perturbation clouds) is close in form to the training objective; we therefore demote it to a mechanism check and base the main claims on Tier 1–6 (Section 4).

BGFM++ (future work). The current model is a pure flow generator whose density is shaped at training time. A natural extension is BGFM++, which adds an explicit scalar energy head $E_\psi(x)$ to the FlowMol3 backbone, calibrated against OMol25 and a held-out xTB/DFT subset, and applies a short MALA / Langevin refinement step after flow generation. This combines BGFM’s amortized flow sampling with EBMol-style explicit energy ranking, and is intended to address EBMol’s high-NFE sampling cost while inheriting flow’s diversity. We sketch the architecture in Appendix D.

6 CONCLUSION

BGFM proposes a training-time use of universal neural potentials as a physical regularizer for de novo 3D molecular flow matching. We do not introduce a new neural backbone; we change the training objective. The key methodological observation is that within-parent variance of $\log p_\theta + E/kT$ removes the unknown $\log Z_c$ degeneracy and lets us embed an FFJORD log-density estimator into FM training at controlled cost. We hope this opens a route to physics-aware de novo molecular generation across the chemistry covered by universal NPs.

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A DERIVATIONS

A.1 SCORE FROM FM VELOCITY FOR LINEAR INTERPOLANT

For the linear interpolant $x_t = (1-t)x_0 + tx_1$ with $x_0 \sim \mathcal{N}(0, \sigma^2 I)$ and $x_1 \sim p_{\text{data}}$, the conditional Gaussian gives $x_0 | x_t, x_1 \sim \mathcal{N}(x_t - tx_1, (1-t)^2 \sigma^2 I)$. Marginalizing, $x_0 | x_t \sim \mathcal{N}(x_t - t\mathbb{E}[x_1 | x_t], (1-t)^2 \sigma^2 I)$. The score of the marginal $p_t(x_t)$ satisfies $\nabla_{x_t} \log p_t(x_t) = -\mathbb{E}[x_0 | x_t] / ((1-t)\sigma^2) \cdot \sigma$, which after substituting $v^* = \mathbb{E}[x_1 - x_0 | x_t]$ and simplifying gives $s(x_t, t) = (tv^* - x_t) / ((1-t)\sigma^2)$. Substituting v_θ for v^* yields equation 2.

A.2 VARIANCE LOSS INVARIANCE TO PER-MOLECULE OFFSET

For any r -independent shift c_m , $\text{Var}_k(y_m^{(k)} + c_m) = \text{Var}_k(y_m^{(k)})$ by translation invariance of variance. Setting $c_m = -\log Z_{c_m}$ shows that the Boltzmann constancy condition $\log p_\theta^{\text{pos}} + \beta E_{\text{NP}} = -\log Z_{c_m}$ is exactly the minimizer of equation 4.

A.3 DIMENSIONAL ANALYSIS OF λ VALUES

The force loss equation 3 is dimensionless (cosine). The energy loss equation 4 scales as $\text{Var}_k(\log p + E/kT) \sim (E/kT)^2 \sim 1/kT^2$ at peak violation. At room temperature ($kT = 0.025$ eV), $1/kT^2 \sim 1600$ times larger than at $kT = 1$ eV; hence λ_2 at room T should be $1/1600$ of high-T. The anchor loss scales as $(\log p + E/kT)^2 \sim 10^{10} - 10^{11}$ at room T after clamping $|\log p_\theta|$ at 10^6 , so $\lambda_3 = 10^{-10}$ keeps the anchor contribution comparable in magnitude to $\mathcal{L}_{\text{FM}}$.

B HYPERPARAMETERS AND REPRODUCIBILITY

Architecture. GVP-Transformer from FlowMol3 (Stark et al., 2024): $n_{\text{hidden_scalars}} = 256$, $n_{\text{hidden_edge}} = 128$, $n_{\text{vec_channels}} = 32$, $n_{\text{molecule_updates}} = 6$, $n_{\text{recycles}} = 1$, total ~ 5.9 M parameters.

Optimization. AdamW with $\text{lr} = 2 \times 10^{-4}$, weight-decay 10^{-12} , warmup-linear 0.02 epoch fraction. Batch size 16 (gradient accumulation 2). bf16-mixed precision with NaN-skip on gradient. Max edges per batch 4×10^4 , max atoms per molecule 200.

BGFM peak hyperparameters (room T). $\lambda_1 = 0.05$, $\lambda_2 = 10^{-4}$, $\lambda_3 = 10^{-10}$, $t_{\text{eval}} = \{0.7, 0.8, 0.9\}$, $K_{\text{ODE}} = 4$ for energy loss, $H = 4$ Hutchinson samples, energy-loss frequency $= 1/4$ steps. Perturbation $\sigma_p \in \{0.03, 0.06, 0.10, 0.20, 0.40\}$ Å, $K = 5$ shards.

Hardware. One NVIDIA H200 141GB GPU per training run, ~ 45 hours to reach 10^5 optimizer steps. Multi-seed runs are embarrassingly parallel across nodes.

C BUG FIXES DOCUMENTED IN THE RELEASED CODE

An external code-and-paper audit (2026-06-29) surfaced three real bugs and several over-claims; this submission incorporates the fixes.

B1 Eval hardcoded charge = 0, spin = 1. The Stage-1 eval (`scripts/eval_boltzmann_stage1.py`) wrote `"charge": 0` into every output record. Stage 2 then queried OMol25 with this charge. We measured the OMol25 4M val charge distribution: 56.22% neutral, 17.91% +1, 12.02% -1, and so on, with 43.78% of held-out molecules carrying nonzero total charge. The patch recovers true charge from `val.atom_charges[ns:ne].sum()`. Stage 1 R^2 numbers obtained before this fix should be treated as a lower bound for charge-sensitive slices.

B2 T-conditional kT not passed to FFFJORD energy loss. In the temperature-conditional variant, the force forward received the sampled kT but the energy-consistency FFFJORD calls did not. The patch plumbs the same kT tensor into `log_density_via_flow` and into `energy_consistency_loss_per_mol(with_anchor)` so that v8b multi-temperature training is correctly conditional.

B3 Eval did not support T-conditional checkpoints. The Stage-1 eval erased the `bgfm` config block before model construction, so the zero-init kT projection module was not installed and T-conditional checkpoints could not be evaluated. The patch adds a `--kT` CLI argument, installs the projection before `load_state_dict`, and passes kT into FFFJORD density evaluation.

In addition to these three bugs, the patch cleans several over-claiming docstrings (the score derivation now reads as a finished proof rather than as "DOUBLE-CHECK"; the anchor is described as an offset stabilizer rather than a partition-function predictor; the Stage-2 print no longer says "learned density IS Boltzmann").

D BGFM++ SKETCH: FLOW PROPOSAL + CALIBRATED ENERGY REFINEMENT

The base BGFM model shapes the implicit density at training time but relies entirely on the flow ODE for sampling. A natural extension is BGFM++, which adds an explicit scalar energy head $E_\psi(x)$ and a short Langevin / MALA refinement after flow generation:

1. Sample $r_0 \sim p_0$, run the trained BGFM flow for $K_{\text{flow}} \in [100, 500]$ steps to obtain a candidate molecule x .
2. Apply $N_{\text{ref}} \in [20, 100]$ steps of MALA / Langevin refinement using $E_\psi(x)$: $r \leftarrow r - \eta \nabla_r E_\psi(r) + \sqrt{2\eta kT} \xi$, $\xi \sim \mathcal{N}(0, I)$.
3. Accept by a Metropolis step against $\exp(-E_\psi/kT)$ to keep the chain valid; optionally light-filter by OMol25 if budget allows.

E_ψ is trained jointly with the BGFM flow, with three calibration signals: regression onto OMol25 energy on training samples, a ranking loss matching xTB ΔE_{relax} on a 200-molecule held-out subset, and a small DFT subset for single-point calibration. We expect this design to recover EBMol-level physical quality at substantially lower total NFE because the flow proposal already places samples near the data manifold; only short refinement is needed near it.

For honest accounting, all reported NFEs would include both the flow ODE steps and the refinement Langevin steps, and all wall-clock numbers would include the energy-head forward passes.